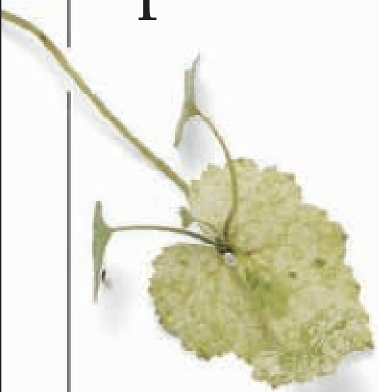
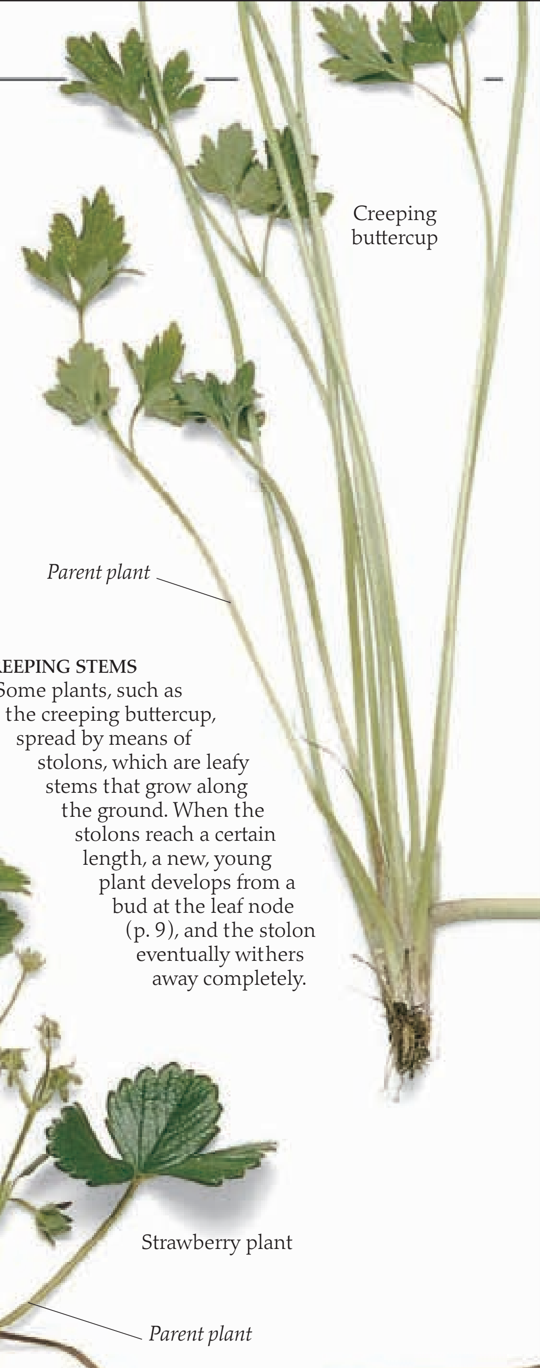
Spreading without seeds

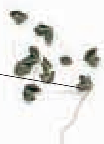


THE PIGGYBACK PLANT
The piggyback plant has an unusual way of reproducing. Tiny new plantlets grow at the base of the older leaves and look as though they are having a piggyback ride.

PLANTS CAN REPRODUCE in two very different ways. In addition to reproducing by means of seeds, they can sometimes also turn small pieces of themselves into new plants. This is known as vegetative reproduction. When a plant reproduces in this way, the young plantlets are genetically identical to the parent. This is very different from reproduction with seeds, which produces seedlings that are all slightly different from their parents. Vegetative reproduction is very useful for farmers and gardeners. It allows them to multiply a plant that has attractive flowers or tasty fruit, knowing that each young plant will have exactly the characteristics they want. The world's oldest plants all grow by forming clumps—another form of vegetative reproduction. In the southwestern United States, there are clumps of creosote bushes up to 12,000 years old, which makes them twice as old as the oldest single trees. In Tasmania, Australia, some clumps of king's holly are more than 40,000 years old, and still growing strong.



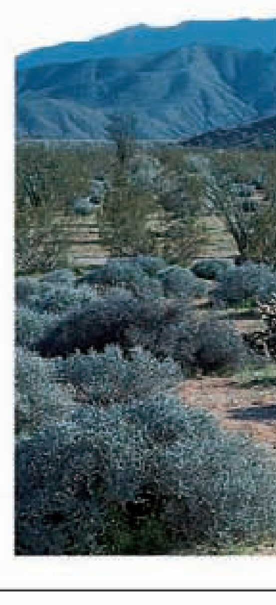
CREEPING STEMS
Some plants, such as the creeping buttercup, spread by means of stolons, which are leafy stems that grow along the ground. When the stolons reach a certain length, a new, young plant develops from a bud at the leaf node (p. 9), and the stolon eventually withers away completely.



Chandelier plant

PLANTLETS AT LEAF TIPS
Kalanchoes are succulents (p. 53), many of which reproduce by developing tiny plantlets along the edges of their leaves. Others, like this chandelier plant, have them just at the tips. When a plantlet is mature, it falls off the parent plant and takes root in the soil beneath.

STRAWBERRY RUNNERS
After they have fruited, strawberry plants produce long stolons known as runners, which spread out over the ground. Strawberry growers wait until the young plants have rooted and then cut the runners. The new plants can be transplanted to make a new strawberry bed.



A MYTH EXPLODED
The famous tumbleweed of the North American prairies is uprooted by strong winds after it has flowered and is often blown far away from the place where it grew. The dead plant cannot put down roots once it comes to a halt, as is often supposed. Instead, it spreads by seeds. The plant scatters them as it tumbles along the ground.